ATOM COUNT CARD:REFINED SUGAR

A.PRODUCT IDENTITY



SWEET CRYSTAL REFINED SUGAR

COMPOUND : SUCROSE, $C_{12}H_{22}O_{11}$

WHY SUCROSE IS IN THE PRODUCT (THE SWEETENER)

B.BASE DATA

Item	Value	Source	Sig figs
Net weight	250 g	Label	2
Package basis	1 pack = 250 g	Label	
Molar mass	342.30 g/mol	PubChem	Not limiting
Formula	$C_{12}H_{22}O_{11}$	PubChem	Exact
Atomic masses	C 12.011, H 1.008, O 15.999 g/mol	CIAAW (2024)	Not limiting

C.MOLAR MASS

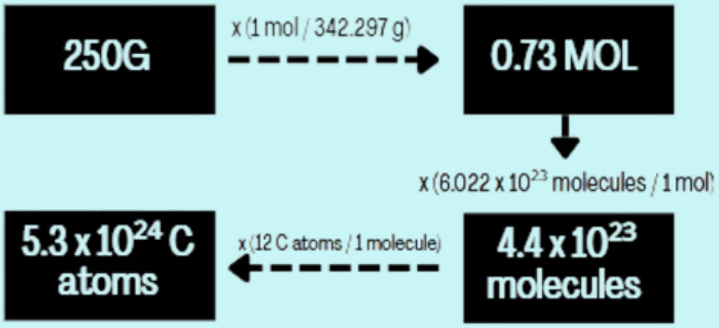
C: 12 × 12.011 = 144.132 (42.107 %)

H: 22 × 1.008 = 22.176 (6.4786 %)

O: 11 × 15.999 = 175.989 (51.414 %)

TOTAL: 342.297 G/MOL

D.MOLE MAP



E. INSIDE THE FORMULA

ATOMS OF EACH ELEMENT

C: 4.3983 × 10^{23} MOLECULES × (12 C ATOMS / 1 MOLECULE) = 5.278 × 10^{24} → 5.3 × 10^{24} C ATOMS

H: 4.3983 × 10^{23} MOLECULES × (22 H ATOMS / 1 MOLECULE) = 9.676 × 10^{24} → 9.7 × 10^{24} H ATOMS

O: 4.3983 × 10^{23} MOLECULES × (11 O ATOMS / 1 MOLECULE) = 4.838 × 10^{24} → 4.8 × 10^{24} O ATOMS

PERCENT COMPOSITION

%C = 144.132 / 342.297 × 100 = 42.107 %

%H = 22.176 / 342.297 × 100 = 6.4786 %

%O = 175.989 / 342.297 × 100 = 51.414 %

TWO ROUTE CHECK

ROUTE A	ROUTE B	ANSWER
C 250 g × 0.42107 = 105.27 g	0.73036 mol × 12 × 12.011 g = 105.27 g	1.1 × 10^2 g
H 250 g × 0.064786 = 16.197 g	0.73036 mol × 22 × 1.008 g = 16.197 g	1.6 × 10^1 g
O 250 g × 0.51414 = 128.54 g	0.73036 mol × 11 × 15.999 g = 128.54 g	1.3 × 10^2 g

REVERSE TO THE FORMULA (100G)

C: 42.107 ÷ 12.011 = 3.5058 MOL · H: 6.4786 ÷ 1.008 = 6.4272 MOL · O: 51.414 ÷ 15.999 = 3.2137 MOL

÷ 3.2137 → C 1.091 : H 2.000 : O 1.000 → × 11 → 12 : 22 : 11

EMPIRICAL FORMULA $C_{12}H_{22}O_{11}$; N = 342.297 ÷ 342.297 = 1

F. SCALE UP

ONE PACKAGE

250 G × (1 MOL / 342.297 G) = 0.73036 MOL → 0.73 MOL SUCROSE

A CLASS OF 36, ONE PACK EACH

36 PACKS × (0.73036 MOL / 1 PACK) = 26.293 MOL → 26 MOL SUCROSE

PACKAGE PER MOLE

1.00 MOL × (342.297 G / 1 MOL) × (1 PACK / 250 G) = 1.369 PACKS → 1.4 PACKS PER MOLE

G. SIG-FIG CHECK AND ONE-NUMBER TAKEAWAY

CALCULATOR VALUE BESIDE REPORTED VALUE

CALCULATOR: 4.398 329 × 10^{23} MOLECULES

REPORTED: 4.4 × 10^{23} MOLECULES OF SUCROSE

LIMIT: THE LABEL'S "250 G" (2 S.F.)

EXACT: AVOGADRO CONSTANT, FORMULA SUBSCRIPTS, 36 PACKS

THE ZERO IN "250 G" IS TREATED AS NOT SIGNIFICANT: 2.5 × 10^2 G

GUARD DIGITS WERE KEPT UNTIL THE LAST LINE.

ONE NUMBER TAKEAWAY

HYDROGEN MAKES UP ALMOST HALF OF THE ATOMS IN SUGAR (22 OF 45), BUT ONLY 6.479 % OF ITS MASS.

H. REFERENCES AND AI DISCLOSURE

SWEET CRYSTAL. (N.D.). REFINED SUGAR, 250 G [SWEET CRYSTAL]. PHOTOGRAPH BY [JOSEF CUBA], OCTOBER 30, 2026.

NATIONAL CENTER FOR BIOTECHNOLOGY INFORMATION. (N.D.). PUBCHEM COMPOUND SUMMARY FOR CID 5988, SUCROSE. RETRIEVED NOVEMBER 2, 2026, FROM HTTPS://PUBCHEM.NCBI.NLM.NIH.GOV/COMPOUND/SUCROSE

COMMISSION ON ISOTOPIC ABUNDANCES AND ATOMIC WEIGHTS. (2024). ABRIDGED STANDARD ATOMIC WEIGHTS. INTERNATIONAL UNION OF PURE AND APPLIED CHEMISTRY. HTTPS://WWW.CIAAW.ORG/ABRIDGED-ATOMIC-WEIGHTS.HTM

BUREAU INTERNATIONAL DES POIDS ET MESURES. (2019). THE INTERNATIONAL SYSTEM OF UNITS (SI) (9TH ED.). HTTPS://WWW.BIPM.ORG/EN/PUBLICATIONS/SI-BROCHURE

AI USE: AI (CLAUDE) WAS USED TO EXPLAIN THE MOLE CONCEPT AND CALCULATION STEPS, WORK THROUGH AND CHECK THE CALCULATIONS, SUGGEST THE CARD LAYOUT AND SCALE-UP, HELP FIND DATA SOURCES (PUBCHEM, CIAAW), AND CHECK WORDING. ALL LABEL VALUES, CALCULATIONS, REFERENCES, AND FINAL ANSWERS WERE VERIFIED BY ME.